

AARON HO

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EDUCATION

University of Southern California

August 2024 to May 2026

Master of Science in Biomedical Data Analytics | GPA: 3.92

Boston University

August 2018 to May 2022

Bachelor of Arts with Honors in Biochemistry and Molecular Biology, cum laude

- Senior thesis: role of DEPTOR and PRAS40 in the metastasis of Tuberous Sclerosis

PROFESSIONAL EXPERIENCE

Sanofi, Manufacturing Sciences, Analytics, and Technology (MSAT)

January 2026 to June 2026

Data Science and Digital Transformation Co-op, Framingham, MA

- Built a hybrid mechanistic digital twin of the Cerezyme ICB perfusion process, modeling cell metabolism, substrate uptake kinetics, mass transfer, and bioreactor dynamics, with Kalman filtering for real-time state estimation
- Refactored the modeling codebase into a production pipeline: automated data ingestion from Snowflake (pilot lab and manufacturing run data), CI/CD with unit tests and pre-commit guardrails, and a UI wired to versioned releases, replacing an unguarded clone-and-push workflow used across the team
- Designed physics-informed control strategies using PID loops and mechanistic bioreactor models, operating alongside Raman-based PAT instrumentation, to support real-time metabolite monitoring and scale-up

Abbott Laboratories, Cardiac Rhythm Management Division (CRM)

May 2025 to December 2025

Computer Engineering Co-op, Sylmar, CA

- Verified and validated Abbott CRM's first FDA-approved AI algorithm for atrial fibrillation and pause detection in insertable cardiac monitors, supporting adjudication of 100k+ cardiac episodes with 99.4% accuracy
- Built a Python-Arduino testing system with a custom GUI for CRM device validation, enabling remote posture simulation and reducing manual adjustments and human error by more than 50%

Neural Modeling and Brain Interface Lab (Song Lab), USC

January 2025 to May 2025

Research Assistant, Los Angeles, CA

- Built neural data processing workflows with Kilosort4 spike sorting for large-scale electrophysiology recordings, developing preprocessing and QC tools for noise filtering, SNR evaluation, and spike train validation

Kwiatkowski Lab, Brigham and Women's Hospital

August 2020 to May 2022

Research Assistant, Boston, MA

- Executed core molecular biology and biochemistry workflows, including PCR, gel electrophoresis, ELISA, and chromatographic protein separation, while maintaining TSC-deficient mammalian cell lines under strict aseptic technique to investigate mTORC1 signaling regulation
- Maintained meticulous, reproducible experimental records and contributed to shared lab protocols, troubleshooting assay conditions and presenting findings to a collaborative research team; culminated in a senior honors thesis characterizing the roles of the mTOR inhibitory proteins DEPTOR and PRAS40 in tuberous sclerosis

PROJECTS

Heart Disease: Multi-Site Clinical ML Study

- Tested whether heart disease risk models transfer across hospitals using leave-one-hospital-out external validation across four cohorts (n=920), with calibration curves, decision curve analysis, and conformal prediction intervals
- Ran a fairness audit across demographic subgroups and characterized where single-site models degrade under distribution shift

GutSense | MIT Grand Hack, 3rd Place (InterSystems Challenge)

- Designed and built an end-to-end colorectal cancer screening platform with an 8-biomarker neural network risk scoring engine deployed on a toilet-mounted biosensor system for continuous, passive, non-invasive monitoring
- Integrated InterSystems IRIS as the health data backbone to ingest patient EHR and medical history, enabling a vector RAG pipeline for contextually personalized risk analysis and AI-generated clinical narratives via the Claude API

Neural Decoding of Imagined Handwriting

- Implemented and compared neural decoding architectures on the Willett et al. intracortical handwriting dataset, including Poisson and Gaussian HMM alignment, an RCNN sequence decoder, and CTC-based decoding

TECHNICAL SKILLS

- Python (NumPy, Pandas, SciPy, scikit-learn, PyTorch, JAX/Flax, Hugging Face Transformers, FastAPI); SQL; MATLAB/Simulink
- Machine learning and deep learning (CNN, ResNet, U-Net, RCNN, Transformers, HMMs); model evaluation: cross-validation, calibration, external validation, conformal prediction, fairness auditing; model serving and inference
- State-space modeling, probabilistic modeling, Kalman filtering, MPC, mechanistic modeling, signal processing (EEG and neural)
- Research software engineering: Git, CI/CD (GitHub Actions), pytest, pre-commit, Ruff, mypy, Hydra, Weights and Biases, Docker, Snowflake